# Does the Type of Disturbance Matter When Restoring Disturbance-Dependent Grasslands?

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## Abstract

**The reintroduction of burning is usually viewed as critical for grassland restoration; but its ecological necessity is often untested. On the one hand, fire may be irreplaceable because it suppresses dominant competitors, eliminates litter, and modifies resource availability. On the other hand, its impacts could be mimicked by other disturbances such as mowing or weeding that suppress dominants but without the risks sometimes associated with burning. Using a 5-year field experiment in a degraded oak savanna, we tested the impacts of fire, cutting and raking, and weeding on two factors critical for restoration: controlling dominant invasive grasses and increasing subordinate native flora. We manipulated the season of treatment application and used sites with different soil depths because both factors influence fire behavior. We found no significant difference among the treatments—all were similarly effective at suppressing exotics and increasing native plant growth. This occurred because light is the primary limiting resource for many native species and each treatment increased its availability. The effectiveness of disturbance for restoration depended more on the timing of application and site factors than on the type of treatment used. Summer disturbances occurred near their reproductive peak of the exotics, so their mortality approached 100%. Positive responses by native species were significantly greater on shallow soils because these areas had higher native diversity prior to treatment. Although likely not applicable to all disturbance-dependent ecosystems, these results emphasize the importance of testing the effectiveness of alternative restoration treatments prior to their application.**


**Key words: fire, grassland, invasive species, oak savanna, restoration.**

## Introduction

One of the more significant "disturbances" for disturbance-dependent ecosystems is the elimination of repeated perturbation. Fire suppression, the alteration of seasonal flooding regimens, or the elimination of grazing can all drastically alter the composition, structure, and function of such systems (Hill & Keddy 1992; Leach & Givnish 1996; Kellman & Meave 1997; Knapp et al. 1998; Turner et al. 2001; Scheller et al. 2005). The mechanistic cause of these changes is usually assumed to be related to competition (Wilson & Shay 1990; Huston 1994; Brewer 1999). In the absence of disturbance, a small number of species are assumed to capture most of the resources (light, nutrients, moisture), thereby suppressing taxa that, typically, can tolerate perturbation but have a limited ability to persist in its absence. In some cases, these few dominant species can alter ecosystem function permanently such as when grassland is converted to forest following woody plant invasion (Wilson 1998). These changes have resulted in disturbance-dependent ecosystems becoming at risk in many regions of the world (Anderson et al. 1999; Folke et al. 2004; Cremene et al. 2005).

As targets for restoration, the first step is often to reintroduce the former disturbances because they are assumed to be critical for recovery. However, there are three potential problems with this approach, all of which are especially applicable to the reintroduction of fire to formerly pyrogenic grasslands: (1) the former dynamics may not be fully understood because they functioned in ways that are no longer apparent (Collins et al. 1998; Keeley 2000; Keeley & Fotheringham 2001); (2) the former disturbances cannot function as before because of habitat loss, population decline, and community change (Leach & Givnish 1996; Schultz & Crone 2005; Varner et al. 2005); and (3) there may be substantial costs and risks with implementation in settled or managed landscapes. The arguments for burning are often based on the assumption that its effects are irreplaceable (e.g., soil nutrients, seed germination, plant mortality). However, this is not always tested so that other types of disturbances may produce the same outcome (Collins et al. 1998; Van Dyke et al. 2004). Further, there are potential risks with the use of fire. First, there is no guarantee that its effects on native plant and animal species will be fully positive due to the small population sizes typically found in remnant areas (Howe 1994*b*; Reed 1997; Varner et al. 2005). Second, if prescribed fires escape to surrounding areas, as occasionally

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happens, the economic costs can be substantial and the subsequent wariness of the public may prevent further application. Determining whether and how to use fire, therefore, has both ecological and practical relevance and needs to be tested against alternative methods for restoring native grasslands.

We examine the impacts of burning versus two other treatments—cutting and raking and the manual removal of the dominant species—on the restoration of native ground flora in fire-suppressed oak savanna of southwestern British Columbia, Canada. This system regularly burned prior to European settlement (Brown & Hebda 1999). Long-term fire suppression has lead to infilling by trees and shrubs and an accumulation of grass litter that normally would have provided fuel for ground fires (Fig. 1). Although the effects of fire suppression are confounded by habitat loss and plant invasion, fire reintroduction is assumed to improve the demographic performance of native flora in part due to their adaptation to its occurrence. Because of the strong correlation between fire suppression and invasion, it is also hypothesized that fire may effectively control invasive taxa. All remnant areas, however, are embedded in a matrix of rural development so that the use of fire has socioeconomic risk. Further, habitat loss of more than 95% means that burning remnant areas (typically <10 ha) may adversely affect rare plants and animals that were formerly wide ranging but now have extremely small populations (e.g., butterflies). The use of alternative methods, such as grass cutting and raking or the manual removal of dominant grasses, could offer greater flexibility for when disturbances are applied (e.g., at the height of fire season) and greater control over the location of the disturbance (e.g., avoiding rare plants or applications on property boundaries) while eliminating the risks posed by burning.

Using plot-level treatments (burning, cutting, removals) in an 18-ha savanna reserve, we tested the relative effectiveness of the three treatments for (1) limiting the growth and reproduction of dominant exotic grasses and (2) increasing the growth and reproduction of subordinate native species. We determined the impact of the treatments on recruitment by subordinate natives species and on resource availability (light, N, moisture). We varied the season of application (summer, fall) and its location (deep soils, shallow soils) to test whether the results of the treatments are contingent on the conditions under which they are applied (e.g., Zedler & Loucks 1968).

## Methods

The Garry Oak (*Quercus garryana)* savanna ecosystem of southwestern British Columbia is part of a prairie and oak savanna complex that extends south to California via the San Juan Islands and Puget Trough of Washington and the Willamette Valley of Oregon. It is associated with a sub-Mediterranean climate, moderately infertile soils, and a former fire regimen that has been eliminated throughout its range since the onset of European settlement during the nineteenth century (Dunn & Ewing 1997;

Figure 1. Invaded (left) and uninvaded (right) oak savanna of southwestern British Columbia. Invaded areas are dominated by exotic grasses that create a dense litter layer in the absence of fire. The uninvaded savanna, reminiscent of the community created by burning, is one of the few remaining deeper soil areas not dominated by invasive grasses. Dominant species are perennial forbs that form a shorter canopy with little persistent litter.

Boyd 1999; Maret & Wilson 2000; MacDougall et al. 2004). Habitat loss and plant invasion affect all savanna remnants from British Columbia to Oregon. It is one of Canada's most threatened terrestrial ecosystems with more than 100 listed "at-risk" plant and animal species (Fuchs 2001).

The experiment was conducted in two grass-dominated sites within the Cowichan Garry Oak reserve in the Cowichan Valley of southwestern British Columbia, Canada (lat 48°48′N, long 123°38′W). The two sites, spaced approximately 300 m apart, vary in soil depth ($\bar{X}$ 42.1 cm [range 21–85 cm] vs. $\bar{X}$ 8.1 cm [range 4.6–18.6 cm]) and species composition (MacDougall 2005). Two perennial exotic grasses (*Poa pratensis* and *Dactylis glomerata*) dominate both sites. The deep soil site contains mostly perennial grasses, annual exotic pasture species (*Vicia* spp. and *Cerastium* spp.), and native forbs that are nonobligate oak savanna species (*Sanicula crassicaulis* and *Osmorhiza berteroi*) (obligate = confined to savanna habitats in British Columbia). The shallow soil site has more native obligate forbs (e.g., *Camassia quamash*, *Dodecatheon hendersonii*, *Ranunculus occidentalis*, *Ce. arvense*, and *Saxifraga integrifolia*). *Camassia quamash* was the only obligate forb species to occur in plots at both sites. The reserve was periodically grazed from the late 1800s to the early 1980s but never cultivated.

The experiment was a randomized complete block design, with no within-block replication. In May 2000, ten 4 × 10–m blocks were located in each site beyond the drip line of the surrounding oak canopy. Plots (each 1 $m^2$) within each block were randomly assigned to one of three treatments applied in July or early October (prior to the rainy season): selective removal of *Poa* and *Dactylis*, cutting and raking, or burning. There was one control plot per block (7 treatments × 2 sites × 10 blocks = 140 plots). The treatments were applied annually from 2000 to 2004. Burning was initiated with a roofing torch and then reapplied to unburned areas to ensure homogeneity of the application. This allowed us to test the impacts of fire in each plot without the confounding effects of unburned patches on postfire vegetation response. An additional 0.25-m buffer was burned around all plots to eliminate overhanging vegetation. Grass was cut to ground level, and all cut material was raked and removed. Weeding removed all aboveground biomass of *Poa* and *Dactylis*. The July treatments had three effects: (1) the removal of the litter layer; (2) the elimination of living foliage of *Poa* and *Dactylis* (weeding) or all species (cutting, burning); and (3) the exposure of the soil surface to intense solar radiation during the summer drought period associated with the sub-Mediterranean climate. The early October treatments eliminated litter and whatever living foliage remained by that time of year (the growing season for the dominant grasses typically extends from the onset of the winter rainy season [November] to mid to late summer).

In May 2000, prior to treatment application, all plots were assessed for (1) species richness; (2) species abundance (cover); (3) species evenness ($E_{var}$; Smith & Wilson 1996); (4) reproductive output of the most common perennial grasses and forbs (number of flowering heads); and (5) light at ground level, soil moisture, bare soil, available soil nitrogen, and soil organic matter (OM). Species abundance and bare soil were visually estimated using a 1-$m^2$ frame divided into 20 cells. Ground-level light (percentage of full light) was measured monthly with a Licor quantum sensor. Soil moisture availability to 12 cm depth (volumetric water content) was measured with a Hydrosense TDR meter. Soil moisture levels were monitored at monthly intervals from March to August and at every 2 months for the rest of the year. Available soil N (mg/kg soil) was determined from 10-cm-deep samples in half of the plots, extracted with 1 mol/L KCL, and analyzed for extractable $NO_3$ and $NH_4$. Percent soil OM was determined by heating 20 g of soil at 400°C for 5 hours and contrasting pre- and postheating soil weight. Most of the variables (e.g., cover, richness, flowering) were remeasured in May of each year from 2001 to 2004. Soil $NO_3$, $NH_4$, and OM were measured three times: prior to the treatments, 1 month after the first application, and 2 years later. In 2000, the fires were generally characterized by measuring flame height; burn duration; and temperatures at, above (50 cm), and below (5 cm) the soil surface, as described in MacDougall (2005). Flame height was less than or equal to 1 m; the burns lasted for less than 10 seconds; and temperature maximums ranged from 133 to 408°C ($\bar{X}$= 263.5°C; $n = 8$ plots). Temperature changes were not detected below the soil surface.

Repeated measures analyses of variance (rANOVA) were used to analyze changes in cover, production, and microenvironmental variability associated with each treatment (Potvin et al. 1990; Kuehl 1994). Because most of the rANOVA tested significantly for sphericity (Mauchly's test), levels of significance were adjusted using Huynh–Feldt and Greenhouse–Geisser corrections. These corrections are more conservative than the unadjusted probabilities but overcome the sphericity problem (Von Ende 2001). In all cases, these two corrections were in agreement on significance. Results of the rANOVA were assessed with Tukey's multiple comparison tests (significance level = $p < 0.05$). All analyses were conducted using JMPIN (SAS 2001).

## Results

Pre-treatment light levels in the grass understory averaged less than 5% of full light at both sites. Deeper soil plots had significantly lower light, averaging 2.14% (SE = 0.37) compared to 3.56% (SE = 0.36) in plots on shallower soil ($F = 2.93$, $p = 0.005$). Deeper soil plots also had significantly greater soil moisture from April to June ($F = 4.33$, $p = 0.0018$; Tukey's HSD); moisture values did not differ between sites in the other months. Percent soil OM averaged 14.9% (SE = 0.48) and did not differ between sites. Levels of available N averaged 192.8 mg kg soil$^{-1}$ plot$^{-1}$ (SE = 94.4) at the deep soil site and 108.2 mg kg soil$^{-1}$ plot$^{-1}$ (SE = 35.6) at the shallow soil site. These differences were not significant ($F = 2.32$, $p = 0.135$).

There were no significant differences in the impact of the three disturbance treatments on ground-level light, available N, soil moisture, and bare soil within each of the two application seasons (Tukey's HSD). After 4 years, each treatment significantly increased light availability and bare soil compared to the control plots but had no effect on soil N, soil OM, or soil moisture (Tukey's HSD). Light levels in spring ranged from 81 to 86% full light among the three July treatments and 62 to 71% for the early October treatments. The disturbance treatments all significantly reduced the growth and reproduction of the dominant grasses (Figs. 2 & 3) and all increased native plant cover and flowering (data only shown for *Camassia* in Figs. 2 & 3), although there were significant interactions with the season of application and site (Table 1).

Summer burning and cutting reduced the cover of the two dominant species to levels similar to those observed in the summer weeding treatment. After 4 years, the average cover of *Poa* and *Dactylis* was reduced to less than 2.5% and less than 2.4% per plot, respectively, at both sites (Tukey's HSD) (Fig. 2). Reproductive effort by both species also declined significantly (Fig. 3). The morphologically similar native $C_3$ grass *Bromus carinatus* responded the same way to burning and cutting, declining significantly in growth and reproduction (Figs. 2 & 3).

The impact of fall cutting and burning did not significantly reduce the cover of *Dactylis*. *Poa* cover, in contrast, decreased significantly from 65.4 to 10%/plot in deeper soils and 25.3 to 11%/plot in shallower soils (Fig. 2).

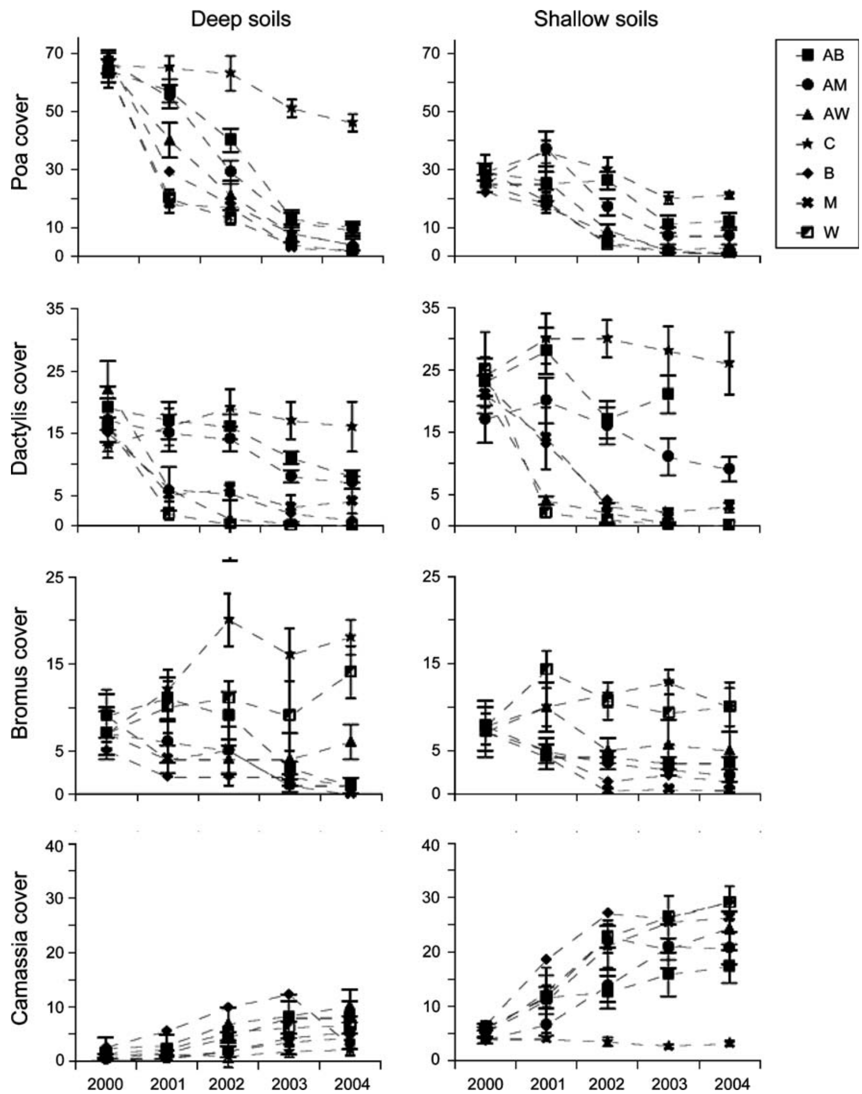


Figure 2. Changes in percent cover for invasive (*Poa* and *Dactylis*) and native (*Bromus carinatus*) perennial grasses and for the most abundant native herbaceous species (*Camassia*) in plots with deep and shallow soils. Treatments were burned (B), mow (M), weeded (W), or control (C), with application in fall (A) or summer. Treatments were applied annually beginning in 2000. Error bars are ± 1 SE.

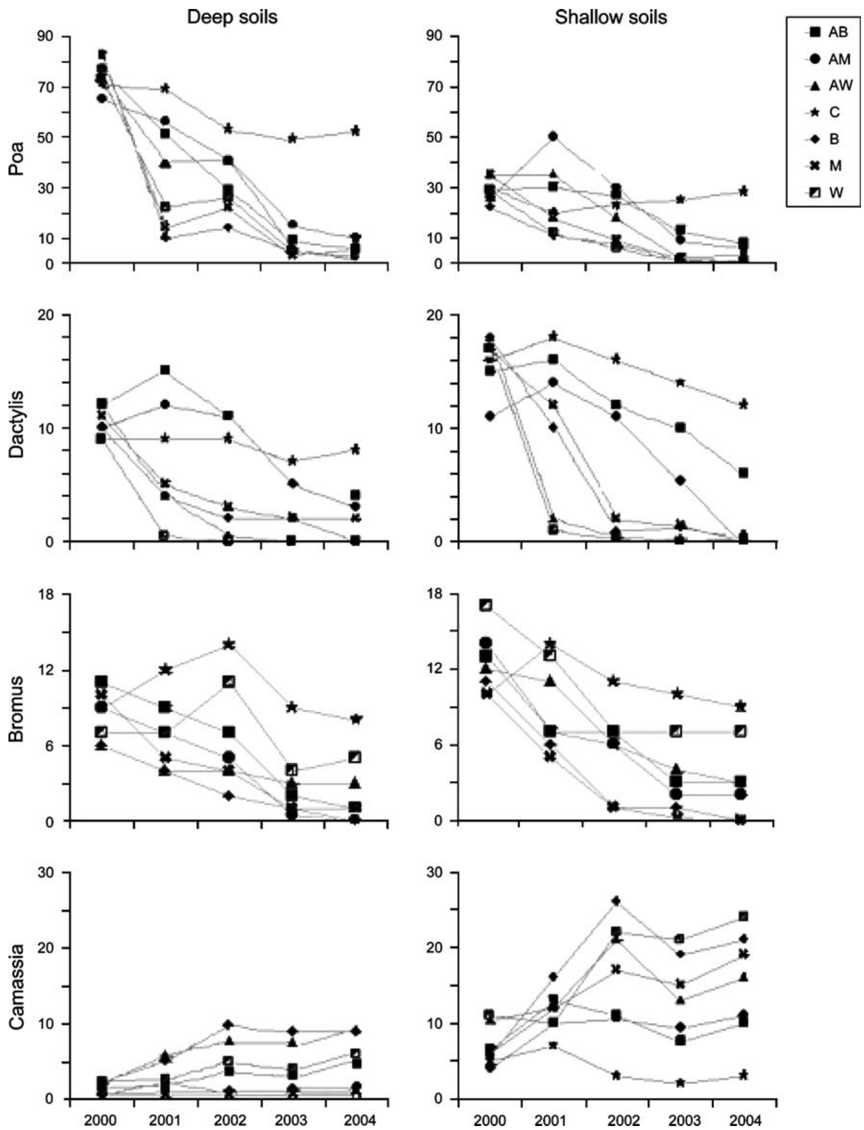


Figure 3. Changes in flowering head production for invasive (*Poa* and *Dactylis*) and native (*Bromus carinatus*) perennial grasses and for the most abundant native herbaceous species (*Camassia*) in plots with deep and shallow soils. Treatments were burned (B), mow (M), weeded (W), or control (C), with application in fall (A) or summer. Treatments were applied annually beginning in 2000. Standard error bars are not shown for sake of clarity.

The treatments caused significant increases in the growth and reproduction of many native plants compared to pre-treatment levels (e.g., *Camassia*; Figs. 2 & 3). Significant increases in cover were observed for native perennial geophytes, especially *Ca. quamash* ($F = 8.44$, $p < 0.0001$), *Dodecatheon hendersonii* ($F = 6.28$, $p < 0.0001$), *Ranunculus occidentalis* ($F = 5.73$, $p < 0.0001$), *Lomatium utriculatum* ($F = 4.61$, $p < 0.0001$), and the nationally threatened *Viola praemorsa* ($F = 2.02$, $p = 0.047$). One exception was the native evergreen perennial forb *Cerastium arvense* that significantly declined in cover ($F = 1.84$, $p = 0.03$) and was eliminated from some plots. All the positive treatment responses for native forbs were confined to plots where these species were already established prior to the experiment. No new recruitment and therefore no growth increases were observed in any previously unoccupied plot, despite elevated levels of light and bare soil from 2001 to 2003.

The positive response of many native species to disturbance unfolded incrementally over the 5 years of the study, as indicated by the highly significant impact of "year" in the analysis of variance model (Table 1). Significant increases in cover, compared to pre-treatment levels, were detectable after 2 years of application for many taxa (e.g., MacDougall 2005); but average cover values in the fifth year were significantly greater compared to year 2 (Tukey's HSD) (e.g., *Camassia* cover in Fig. 2).

**Table 1.** Summary of rANOVA for the individual or interacting effects of site, block, application season, treatment, and year on several community-level response variables.

| *Source* | *df* | *Total Species Richness* | *Native Species Richness* | *Evenness ($E_{var}$)* | *Exotic Grass Cover* | *Native Forb Cover* |
|---|---|---|---|---|---|---|
| Main effects | | | | | | |
| Site (fixed) | 1 | **<0.0001** | **<0.0001** | **0.02** | **<0.0001** | **<0.0001** |
| Block (random) | 9 | 0.07 | 0.07 | 0.69 | 0.06 | 0.11 |
| Season (fixed) | 1 | 0.08 | 0.62 | 0.73 | **0.005** | **0.008** |
| Treatment (fixed) | 3 | **<0.0001** | 0.89 | **0.03** | 0.11 | 0.27 |
| Site × season | 1 | 0.46 | 0.43 | 0.93 | 0.29 | 0.07 |
| Site × treatment | 3 | 0.28 | 0.06 | 0.99 | 0.56 | 0.10 |
| Season × treatment | 3 | 0.51 | 0.49 | 0.95 | **0.03** | 0.57 |
| Site × season × treatment | 3 | 0.11 | 0.08 | 0.85 | 0.19 | 0.6 |
| Error 1 | 144 | | | | | |
| Split effects | | | | | | |
| Year (fixed) | 4 | **<0.0001** | **<0.0001** | **<0.0001** | **<0.0001** | **<0.0001** |
| Site × year | 4 | **<0.0001** | **<0.0001** | **<0.0001** | **<0.0001** | **<0.0001** |
| Season × year | 4 | **0.01** | 0.24 | **0.0002** | **<0.005** | **<0.03** |
| Treatment × year | 12 | **<0.0001** | 0.71 | **0.009** | **<0.0001** | **<0.0001** |
| Season × treatment × year | 12 | 0.57 | 0.95 | 0.55 | **0.02** | 0.13 |
| Site × season × year | 4 | 0.93 | 0.98 | 0.58 | 0.56 | 0.30 |
| Site × treatment × year | 12 | 0.49 | 0.72 | 0.56 | 0.44 | **0.001** |
| Site × season × treatment × year | 12 | 0.11 | 0.33 | 0.37 | 0.21 | 0.26 |
| Error 2 | 711 | | | | | |
| Total | 799 | | | | | |

The analysis was conducted as a split-plot design, with year as the split effect. This analysis structure was used separately for each of the analyzed response variables. Bold indicates significance. *df*, degrees of freedom.

Although there were significant changes in relative abundance of some native taxa, the treatments had no substantial impact on the percentage of native species per plot (Fig. 4). After 4 years of treatments, new recruitment by native perennial species was close to nil, except where they were already established. Forb cover and species evenness ($E_{\text{var}}$) increased significantly over time (Fig. 4) in association with the significant declines in perennial grass cover. There was a significant increase in species richness for all treatments (Fig. 4), explained by the proliferation of ruderal species from the seed bank.

## Discussion

In this savanna, where fire has been suppressed for more than 150 years, burning was an effective tool for controlling exotic dominant grasses and restoring native flora. By increasing the cover of perennial forb species, it shifted the plant community more closely toward the compositional profile described by historical records prior to fire suppression (MacDougall et al. 2004). Because the exotic grasses and the functionally similar native species demonstrated such a high sensitivity to fire, it suggests that the present-day dominance by exotic grasses is more a function of the long-term absence of burning than the uniqueness of their traits or competitive ability (MacDougall & Turkington 2004, 2005).

The impacts of burning, however, were matched by cutting and selective weeding, indicating that biomass removal, including the elimination of accumulated dead litter, was the most important factor for restoration, regardless of how it occurred. This was especially true when the disturbances were applied in midsummer prior to seed set by the dominant exotic grasses. Repeated disturbance by fire, therefore, appears to have been a critical component of ecosystem function prior to the 1840s and currently could be an effective tool for restoration; but its impacts can be mimicked by other methods.

The treatments had similar impacts because light appears to be the primary limiting resource for plant growth and reproduction. This result may not be repeatable in grasslands where different resources are limiting (i.e., soil nutrients), so the impacts of different disturbances are not similar. It is also likely to depend on the method of application of the alternative treatments, compared to effects created by burning (e.g., cutting height, degree of litter removal by raking; Maret & Wilson 2005). As is typical of productive grasslands (Foster & Gross 1998; Hartnett & Fay 1998), light availability in this savanna is restricted by a grass canopy that can exceed 1 m in height and forms a dense litter layer. The species most capable of coexisting are exotic forbs associated with mesic pastures (e.g., *Vicia* and *Cerastium*), which have presumably adapted to these conditions (e.g., low light compensation points and vine-like morphologies that aid in light foraging). The native species most common in these swards are large-sized $C_3$ grasses that are functionally similar to the exotic dominants and the nonobligate forbs associated with neighboring coniferous forest where light levels are also limiting.

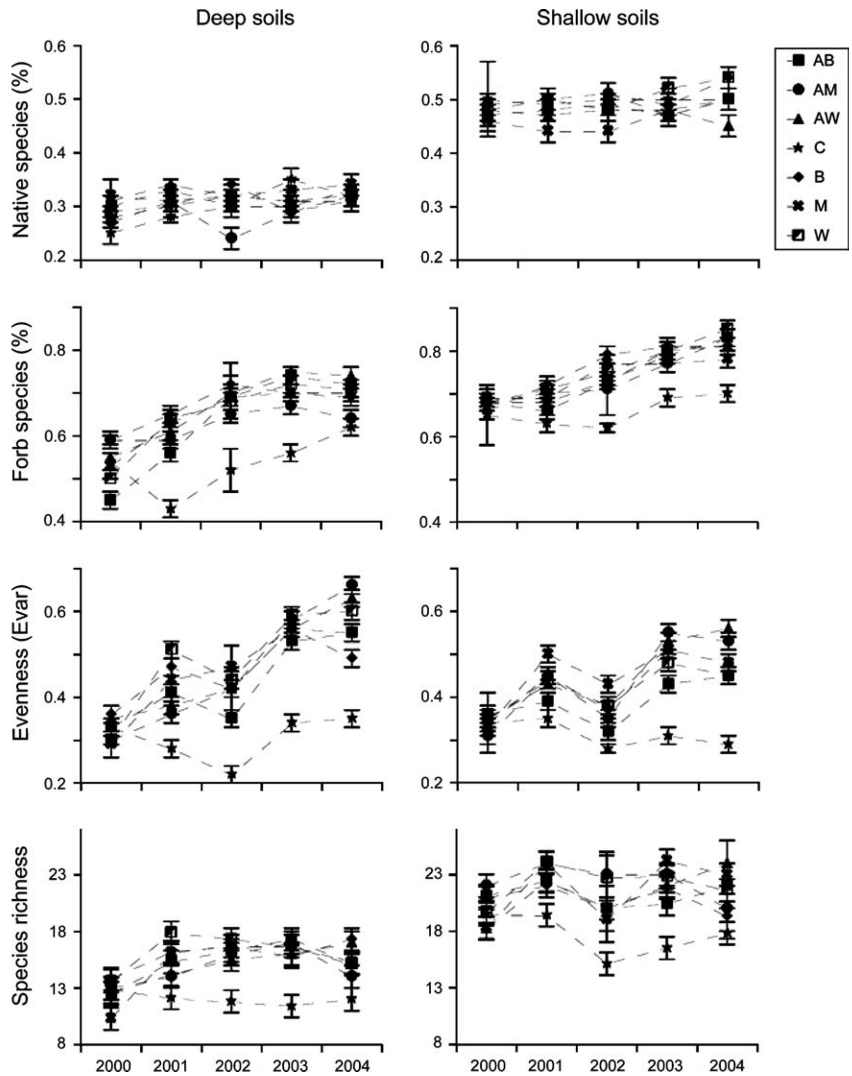


Figure 4. Treatment impacts on native species (% of total species), forb cover, evenness, and species richness in plots with deep and shallow soils. Error bars are ± 1 SE.

We predicted that repeated burning would influence the levels of available soil N, in line with observations from other grassland systems (e.g., Knapp et al. 1998). The effects of fire-available nitrogen could have been positive due to feedbacks between ash deposition, litter quality (lower C:N ratios), and litter quantity (Covington & Sackett 1992; Blair 1997; Knops et al. 2002). Conversely, they could have been negative due to the repeated annual volatilization of the aboveground biomass and reduced OM in the soil (e.g., Blair 1997; Newland & DeLuca 2000; Knops et al. 2002). We found no evidence for either scenario, as no change in available N or OM content was detected, despite substantial changes in species composition. Although unexpected, this result is consistent with other recent studies that have found no net effect of burning or other forms of biomass removal on the soil nitrogen pool (Maron & Jefferies 2001; Wan et al. 2001; Trammell et al. 2004). One possible explanation is that the combined positive and negative effects of fire on N availability offset each other. For example, increases in soil temperatures caused by the removal of aboveground biomass could stimulate microbial activity, but the reduction in litter quality (i.e., higher C:N ratios due to the loss of N by volatilization) could suppress mineralization. We suspect that our results are not a sampling artifact because there were no detectable differences in biomass or litter production by the native flora among the treatments (MacDougall, unpublished data).

In this system, the fire season is thought to have formerly occurred from July to October when moisture availability was lowest due to the sub-Mediterranean climate. Although fires can presumably occur at anytime during this period, our results indicate that the timing of the burn during fire season can significantly determine its impact on the plant community, as has been reported from other grassland systems (Towne & Owensby 1984; Howe 1994*a*, 1994*b*). This result is determined by the phenology of the dominant exotic grasses in the oak savanna. These grasses do not have Mediterranean origins and flower well into the summer dry period. Although the climate is clearly not severe enough to prevent their persistence, it means that they are in full reproductive mode when midsummer fires occur. Disturbing these species at this time is dramatically more effective at controlling their abundance in subsequent years compared to late-season burns.

Unexpectedly, however, these impacts also negatively affected the native $C_3$ perennials. It seems, therefore, that long-term fire suppression has increased the dominance of $C_3$ grasses, regardless of their biogeographical origins. Historical records confirm that tall grass swards occurred naturally in the absence of fire (MacDougall et al. 2004). The timing of fire, therefore, has probably always been important for determining the functional composition of this savanna. Similarly, the absence of fire in some areas of the landscape (spatial refuges) must also have been critical. There is a subset of native species that are evergreen or have late-season phenologies, making them vulnerable to repeated burning (e.g., *Ce. arvense*). Such findings suggest that the timing of the disturbance is important, but it must be matched with the seasonal phenologies of species in the remnant communities. The widespread application of fire in small reserves during midsummer, e.g., may have positive effects for controlling grasses but detrimentally impact late-season plants and animals. This suggests that a spatially explicit model of fire application is required (Hobbs & Huenneke 1992; Collins et al. 1995; Harrison et al. 2003; Knight & Holt 2005), not only targeting exotic species control but also maintaining areas free from intense perturbation.

The degree of positive impact of the treatments on native plants was fully contingent on the composition of the vegetation prior to the start of the experiment. Growth increases were most significant in areas where native flora already occurred, albeit in lower abundances due to the suppressive effect of the dominant grasses (MacDougall 2005; MacDougall & Turkington 2005). In areas lacking native forbs, no increases occurred because there was no new recruitment. This result highlights that the presence or absence of disturbance is unlikely to be the only factor controlling the compositional dynamics of degraded savanna ecosystems. In this savanna, many native species are dispersal limited, as has been reported in other grassland systems (e.g., Maron & Jefferies 2001; Foster & Tilman 2003; Maret & Wilson 2005). Experimental seed additions have shown that the inability to disperse may actually be more limiting for native species than the absence of fire (MacDougall 2005; MacDougall & Turkington 2006). Although it is unclear whether this limitation occurs naturally or is a contemporary phenomenon (MacDougall & Turkington 2005), it indicates that disturbance treatments alone cannot restore native savanna. The more degraded the site, the less likely the recovery by native species other than those already present. Supplemental measures targeting dispersal and the survival of juveniles are needed in conjunction with treatments that reduce the disturbance-sensitive competitive dominants.

The impacts of the treatments on native species continued to unfold for the duration of this experiment, highlighting the importance of maintaining and monitoring restoration programs over periods longer than a few years. The most dramatic initial changes in diversity and plant cover were determined by a flush of annual ruderals resident in the seed bank. The abundance of these taxa peaked in the first year after application and then diminished as slower growing; but competitively superior perennial forbs began increasing in cover (MacDougall 2005). Because these native forbs can often take up to a decade to reach reproductive maturity (e.g., Liliaceae species such as *Camassia quamash, Erythronium oregonum* and *Fritillaria lanceolata*), annual disturbances may be required long into the future to make significant impacts on their demography in areas where they are not already abundant.

## Conservation Implications

By testing the impacts of alternative restoration treatments, we determined that light limitation significantly constrains native plant growth and reproduction in this ecosystem. Current efforts at protecting remnant savanna are increasingly successful; but the creation of protected areas is only the first step. An absence of disturbance in protected areas promotes dominance by exotic grasses that significant limit light availability, so that protection must be coupled with the reintroduction of some form of disturbance. This is necessary to reduce the abundance of exotic grasses and increase the growth, reproduction, and establishment of native species. It also lowers the risk of accidental and severe fires in these reserves, which is not insignificant, given the substantial accumulation of standing litter caused by decades of fire suppression.

Because remnants of this formerly pyrogenic savanna are embedded in a matrix of settlement, farms, and rural development, concern over fire use is high. Our study has shown that the most effective time to burn coincides with the period of highest fire risk. The burns of this study were small sized, ensuring safety; but to be effective as a restoration tool, larger free-ranging fires are essential. Although methods for controlling prescribed fires are well established elsewhere (Packard & Mutel 2005), it can be difficult to conduct large burns in areas where residents are unaccustomed to its occurrence. Our results reveal that there are alternatives to burning when the risks are too high or when fuel quality is too low to support fire (e.g., early-season application). Although we do not suggest that fire is fully replaceable in this system, or in pyrogenic grasslands generally, these results indicate that alternative restoration treatments may have some success for controlling exotics and promoting native species. This suggests the importance of testing the effectiveness of alternative restoration treatments prior to their application.

### Implications for Practice

- No significant difference found between fire, mowing, and weeding on exotic species control or native plant growth in a degraded oak savanna.
- Although we do not suggest that fire is fully replaceable, there may be alternatives to burning in some systems when the risks are high.
- It is important to test the effectiveness of alternative restoration treatments prior to their application.

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